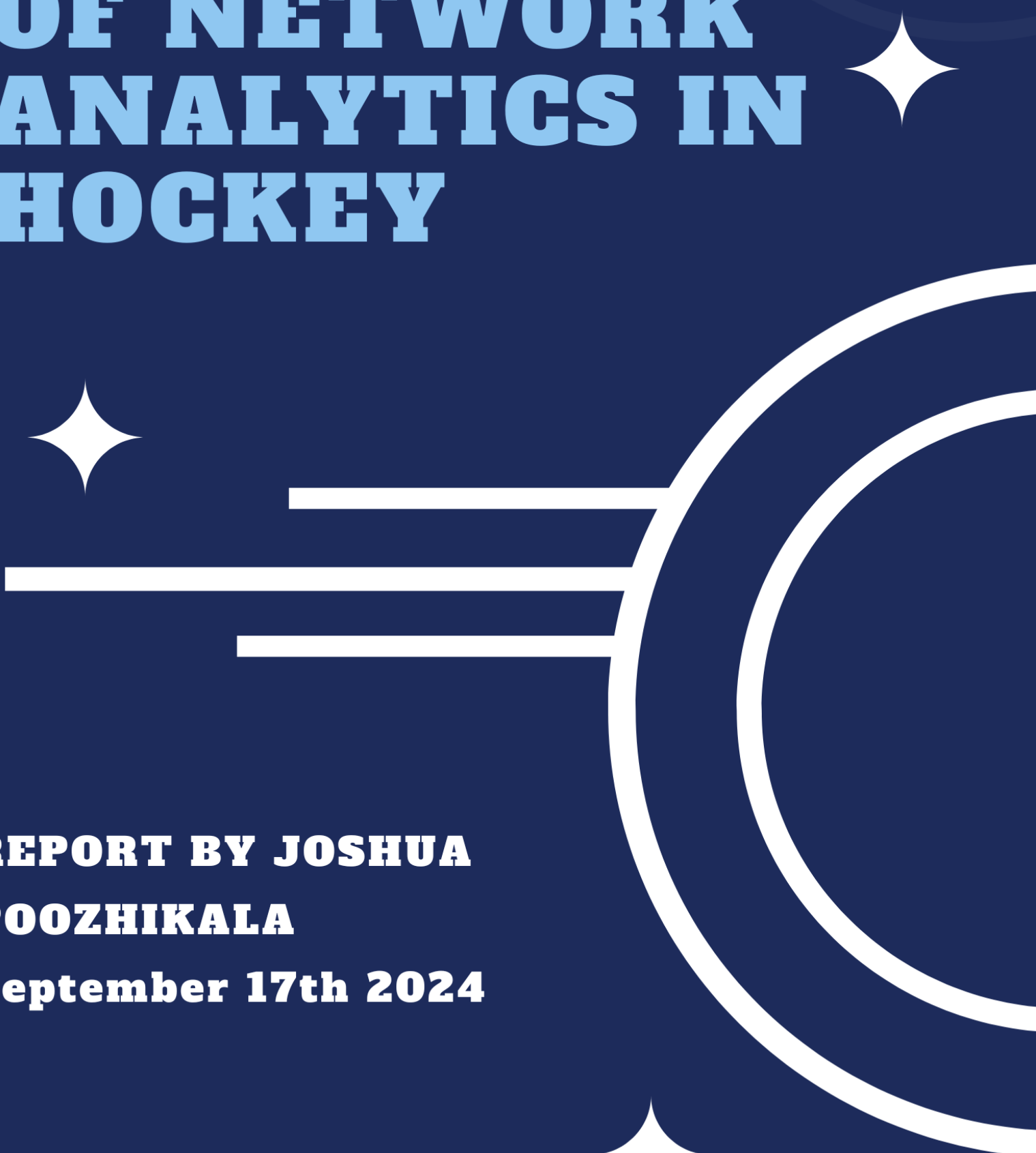


APPLICATIONS OF NETWORK ANALYTICS IN HOCKEY

A stylized white graphic on a dark blue background. It features a hockey stick with three horizontal blades and a large circular puck. Three white four-pointed stars are scattered around the graphic: one in the upper right, one to the left of the stick, and one at the bottom center.

**REPORT BY JOSHUA
POOZHICALA
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INTRODUCTION

Network analysis has evolved into a powerful tool for understanding complex systems by examining relationships between entities, represented as nodes and their connections, represented as edges. Over time it has expanded its applications from social dynamics to fields as diverse as biology, neuroscience, and the internet. The power of the analysis lies in its ability to uncover behavioral patterns and dynamics within networks, as well as provide understanding to the roles of the individuals involved. While network analysis has been applied extensively in sports like soccer, its use in hockey remains underexplored. Like all team sports, success in hockey depends on a network of players performing at their best, both individually and collectively. However, understanding the connections between these individuals can be complex. Analyzing the role of each player within this dynamic process is crucial for gaining a deeper understanding of how a team functions. Traditional metrics in hockey, such as goals, assists, and points, only tell part of the story. To fully grasp a team's performance, it's essential to look beyond these stats and explore the deeper dynamics of player interactions and contributions within the network. By analyzing passing networks, goal sequences, and possession flows, network analysis provides insights into player performance, influence, and team dynamics beyond traditional statistics, making it a valuable method for evaluating players. This report will explore the applications of network analysis in two different contexts within hockey.

1. Evaluating draft prospects to predict their level of success in the NHL
2. Understanding the impact of a player within the context of a singular game.

ANALYZING A PROSPECTS SUCCESS THROUGH NETWORK ANALYSIS

METHODOLOGY

To analyze the correlation between network analysis metrics and the future performance of OHL prospects, a dataset curated by Evan Oppenheimer, a data scientist in the hockey analytics space, was used (See Appendix for link). The dataset includes stats for draft-eligible players in the OHL, QMJHL, WHL, USHL and AHL over the last 25 seasons. The dataset is focused on a metric known as Betweenness Centrality. This measure identifies nodes that are critical for the flow of information within a network. In this context, a node (or player) with high Betweenness Centrality is more integral to the overall network compared to their teammates. This dataset was merged with NHL player data from QuantHockey.com (Refer to Figures 1 and 2 in Appendix). The dataset contained statistics such as shots, goals, assists, points and P/60 Min.

In hockey, a player with high Betweenness Centrality is part of many different networks of players that are involved in scoring and assisting on goals. They are connected to a wide range of teammates, and their involvement is crucial to the success of the team. For example, if Player A frequently passes the puck to Player B, who then passes it to Player C to score, Player B has a high Betweenness Centrality. This is because Player B acts as the key connector in the play, linking Players A and C. If Player B were removed, it would disrupt the connection between A and C, making it harder for them to execute their plays. The betweenness score for each player was derived from goals and assists in game-summary data, however this metric can be misleading in some respects as it does not boost players if they have a strong goal scoring ability. To address this, an Average Rating metric was created by combining points per game and Betweenness Centrality. This adjustment better reflects a player's overall ability as it penalizes those with inflated betweenness scores and low points while rewarding those who contribute more directly to goals via a goal or assist.

FINDINGS AND ANALYSIS

After combining the datasets, several interesting findings emerged. Linear correlations were evident between Average Rating and Even Strength Goals, Points/60 minutes, and Points/Game. Additionally, Betweenness Centrality showed linear correlations with Points/60 minutes. These trends suggest that both Average Rating and Betweenness Centrality provide valuable insights into a player's effectiveness in the NHL. The strong correlations with Points/60 minutes are particularly significant, as this normalized metric compares players with the exclusion of total ice time, giving a clearer understanding of performance on a per-minute basis. The fact that both Average Rating and Betweenness Centrality correlate with this metric highlights their importance in assessing a player's impact. Overall, this demonstrates that players with higher Betweenness Centrality scores and Average Ratings in their junior hockey play are more likely to be effective and efficient in the NHL. This also highlights the applications that Network Analysis has in being a predictor for future career success. See Appendix: Figure 3 for graphs and correlation values.

UNDERSTANDING IN-GAME PERFORMANCE THROUGH NETWORK ANALYSIS

METHODOLOGY

Network analysis can offer deeper insight into a player's impact within a game. While traditional stats like points, assists, and goals provide many insights into performance, they often fail to capture important contextual factors. Using play-by-play data from the first four games of the 2023-2024 Rivalry Series between the Canadian and American

Women's Hockey teams, network analysis algorithms and metrics were applied to enhance the understanding of player performance.

The dataset used includes information about various game events such as faceoff wins, puck recoveries, plays (successful passes), and incomplete plays (unsuccessful passes). For each event, the data records the period and time remaining, the skaters on the ice for each team, goals scored, and the specific players involved in the events. Additional event details specify the nature of the event, such as the pass type (direct or indirect) for plays or the method of puck recovery. (See Appendix: Figure 4)

Using this dataset, I designed a network analysis algorithm that identified and visualized sequences of events that lead to a shot on net or a goal. This was done by first defining the types of events that indicate a team is in possession of the puck, such as "Play," "Puck Recovery," "Faceoff Win," and "Takeaway." The algorithm then identifies sequences of events where a team gains possession and ultimately takes a shot on net or scores a goal. Once these sequences are identified, a Flow Centrality value is assigned to each player involved in the scoring opportunity. These values are then aggregated across all sequences to create a final score for the player. Flow Centrality measures a player's contribution to the overall flow of action by tracking their involvement in sequences, giving greater weight to players who are closer to the final shot or goal. It differs from Betweenness Centrality in that it assigns a weight to each edge of the network based on how close it is to the final objective (a shot on net or goal). This is done because not all passes are considered equal within a sequence of plays; passes closer to the final objective are treated as more critical.

FINDINGS AND ANALYSIS

Figure 5 (See Appendix) depicts all sequences of play that end in a shot on net or lead to a goal, though dense with data, it reveals many valuable insights when explored further, specifically, when looking at the Flow Centrality measures associated with the visualization. Despite scoring 0 goals in four games, Marie-Philip Poulin, had the second highest Flow Centrality score amongst all players. Using network analysis, it was identified that she was involved in 24 different sequences, 10 of which ended with her taking a shot on target. Marie was frequently involved in long passing sequences that led to shots on target. Figure 8 (See Appendix) presents an example of this. The image shows an 8-pass shot-on-goal sequence in which Marie was involved, making 3 of the passes, including the final shot on target. Due to sequences like this, Marie accumulated the second-highest flow centrality score among all players, with a value of 24.97. Despite not recording any points in the first four games, she scored 4 points in the following three. A similar pattern was observed with Brienne Jenner, who had the highest flow centrality score of 31.28 recording only 1 point during the first four games but registering 4 points in the following three. In both cases, their high flow centrality in

the earlier games signaled strong underlying performance, predicting future goals and assists. This suggests that the issue in the first few games was missed scoring opportunities rather than poor play from these players.

Another interesting finding emerged when comparing the networks of Taylor Heise and Alex Carpenter, both forwards for Team USA. Taylor finished with 3 points over the four games, while Alex had 4. At first glance, based on the box score, Alex Carpenter seems to have had a better performance. However, when examining the centrality measures, it becomes clear that Taylor had a similar or even better performance. Taylor's centrality score was 23.91, compared to Alex's 12.07, indicating that Taylor was more integral to creating scoring opportunities. The visualizations (See Appendix Figures 6 & 7) further support this, showing that Taylor's network is denser and contains more connections. The network measures are further supported by other stats. She had double the shots on goal compared to Alex and was involved in 26 scoring opportunities, whereas Alex was involved in 18.

One of the most valuable applications of network analysis within the context of hockey is in quantifying the offensive impact of defensive players. Take Renata Fast, a defender for Canada, as an example. Defenders' offensive contributions are often overlooked. However, they play a crucial role in initiating, generating, and supporting offensive plays. It was revealed that Renata had a flow centrality score of 20.46—one of the highest among defenders and the 8th highest overall. Compared to other defenders, Renata had the second-highest number of shots on target (5). She also completed the second-most passes of any player (130) and led all players in puck recoveries (128). In contrast, when examining Savannah Harmon, a key defender for Team USA, her traditional stats appear like Renata's, with both recording just 1 assist over the four games. However, her centrality measures tell a different story—Savannah's centrality score is significantly lower than Renata's, at 5.89. Her passing network also shows a much smaller and condensed behaviour (See Appendix Figures 9 & 10). These findings are supported by the fact that she only had two shots on goal and completed 73 passes in the four games. Attached with this report is a Tableau visualization highlighting the shot on goal sequences that occurred during the Rivalry Series. A more detailed look at the flow centrality calculations and model output for each play can be seen within the dashboard (See Appendix Figure 11).

IMPROVEMENTS

While the current analysis provides valuable insights into player performance through network analysis, there are several areas for improvement that could enhance its effectiveness in the context of hockey.

One key area for refinement is the weighting system used in flow centrality calculations for shot-on-goal sequences. Currently, the player taking the shot is assigned the highest weight of 1, with a decay factor of 0.8 applied to preceding events in the sequence. This approach emphasizes the direct impact of the shot but may overlook the significant contributions of earlier actions.

To enhance the model, future iterations could explore alternative weighting schemes that provide a more nuanced understanding of game dynamics. For instance, rather than simply assigning a fixed value for taking a shot, a score based on the shot's quality could be given. Additionally, higher weights could be assigned to key plays, such as takeaways, puck recoveries in the opponent's zone or faceoff wins, recognizing their critical role in the buildup to a shot. Overall, adjusting the logic for assigning points within shot-on-goal sequences will help illuminate the more critical plays that contribute to scoring opportunities, leading to deeper insights into player influence at different stages of play.

Additionally, when analyzing flow centrality scores in play-by-play data, introducing an adjusted flow centrality score that accounts for player time on ice would be a crucial improvement. This adjustment would enable more meaningful comparisons between players, offering a better understanding of who is most efficient at generating scoring opportunities during their time on the ice.

Another critical enhancement lies in expanding the dataset used for analysis. Applying this network analysis model to larger, more comprehensive play-by-play datasets—spanning multiple games, seasons, and levels of play—would allow for a more robust evaluation of the model's predictive capabilities. By incorporating this broader dataset, the model could better capture player patterns and performance trends, making it possible to evaluate its effectiveness against traditional methods of player assessment. This would not only improve the model's accuracy but also enable more conclusive comparisons between network analysis and other performance metrics.

CONCLUSION

In conclusion, the paper highlights several use cases of network analysis in evaluating player performance. While this metric should not replace traditional statistics like goals, assists, and points, it serves as a valuable complement by revealing insights that conventional metrics might miss. Network-based metrics, such as betweenness centrality, show promise in identifying player performance trends and correlating them with future outcomes, as demonstrated in the analysis of OHL players and their subsequent NHL careers. The most effective applications, however, appear to be in evaluating player performance over shorter time windows, offering a clearer

understanding of their immediate impact on the game, as seen in the analysis of shot-on-goal sequences in the 2023 Rivalry Series.

APPENDIX

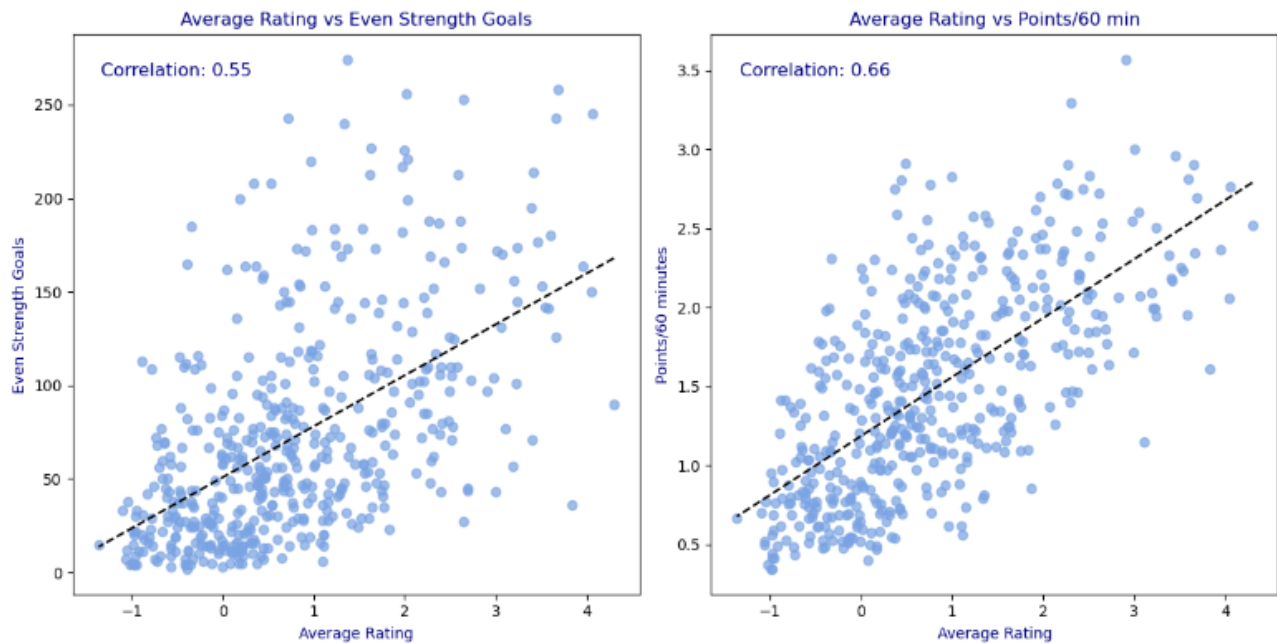
Figure 1: NHL Betweenness and Average Rating Statistics
(<https://medium.com/@evanopp/whose-point-is-it->

name	position	season	league	team	age	gp	betweenness_all_adj	points_per_game_all_adj	avg_all_rating
Sidney Crosby	F	2004-05	QMJHL	RIM	17.109	62	0.1671	2.3017	5.1889
Connor McDavid	F	2014-15	OHL	ER	17.67	47	0.1368	2.0776	4.5382
Kenndal McArdle	F	2004-05	WHL	M.J	17.698	70	0.3304	0.6919	4.5052
Steven Stamkos	F	2007-08	OHL	SAR	17.602	61	0.2332	1.2591	4.3304
Dylan Strome	F	2014-15	OHL	ER	17.525	68	0.2065	1.4257	4.2978
Nikita Scherbak	F	2013-14	WHL	SAS	17.712	65	0.2936	0.7442	4.197

Figure 2: NHL Player Statistics Data over last 25 seasons from Quant Hockey

Rk	Name	ESP	PPP	SHF	GWP	OTP	PPP%	G/60	A/60	P/60	ESG/60	ESA/60	ESP/60	PPG/60	PPA/60	PPP/60	G/OP	A/OP	P/OP	SHOTS	SH%
1	Sidney Crosby	1025	560	11	259	44	35.1%	1.350	2.289	3.639	1.223	1.791	3.014	1.915	4.319	6.233	0.465	0.789	1.255	4072	14.5%
2	Alexander Ovechkin	963	581	6	231	40	37.5%	1.731	1.414	3.145	1.421	1.132	2.553	2.775	2.393	5.168	0.598	0.469	1.087	6627	12.9%
3	Joe Thornton	973	547	12	249	31	35.7%	0.809	2.095	2.904	0.669	1.698	2.368	1.475	4.052	5.528	0.257	0.666	0.923	3035	14.1%
4	Evgeni Malkin	799	492	5	219	40	38.0%	1.327	2.126	3.453	1.089	1.674	2.763	2.189	3.828	6.017	0.435	0.697	1.132	3685	13.5%
5	Patrick Kane	866	417	1	202	39	32.5%	1.139	1.966	3.104	1.014	1.531	2.544	1.746	4.033	5.779	0.383	0.661	1.044	4058	11.6%
6	Jarome Iginla	797	406	15	193	16	33.3%	1.282	1.361	2.643	1.108	1.145	2.253	2.064	2.370	4.434	0.422	0.447	0.869	4436	13.3%
7	Anze Kopitar	761	423	27	199	32	34.9%	0.876	1.656	2.532	0.791	1.321	2.112	1.587	4.008	5.595	0.305	0.577	0.882	3230	13.0%

Figure 3: Correlations for Betweenness Centrality, Average Rating and Stats in NHL



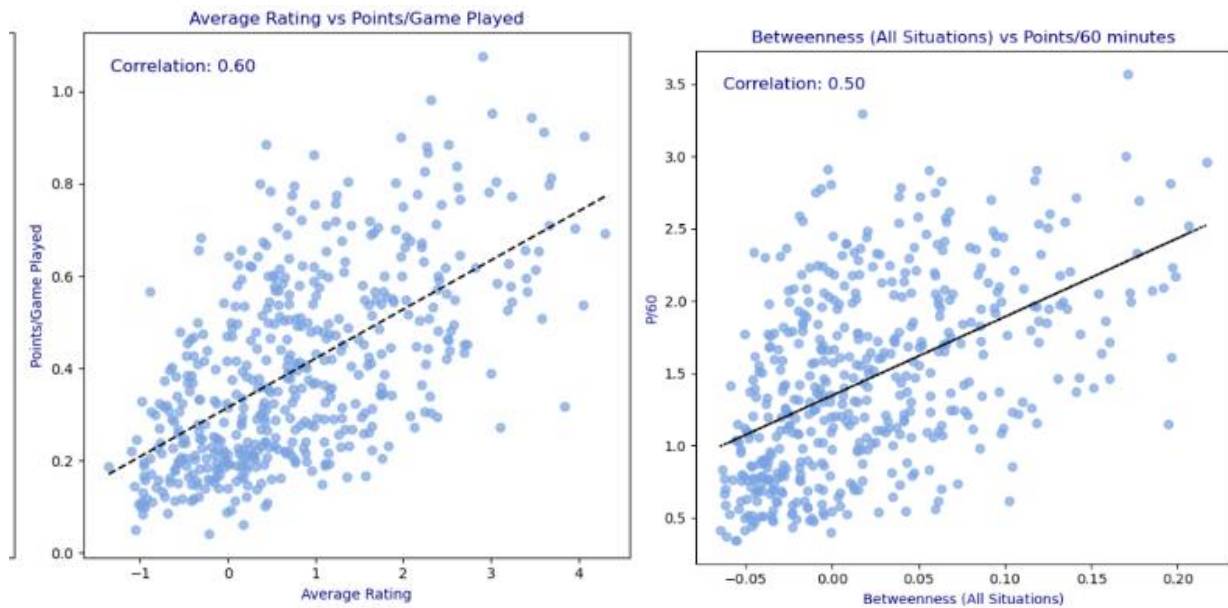


Figure 4: Play-By-Play Dataset from 2023-2024 Rivalry Series: Canada vs USA

Date	Home Team	Away Team	Period	Clock	Home Team Skaters	Away Team Skaters	Home Team Goals	Away Team Goals	Team	Player	Event	Detail 1	Detail 2	Detail 3	Player 2
2023-11-08	Women - United States	Women - Canada	1	20:00	5	5	0	0	Women - Canada	Marie-Philip Poulin	Faceoff Win	Backhand			Taylor Heise
2023-11-08	Women - United States	Women - Canada	1	19:57	5	5	0	0	Women - Canada	Jocelyne Larocque	Puck Recovery				
2023-11-08	Women - United States	Women - Canada	1	19:54	5	5	0	0	Women - Canada	Jocelyne Larocque	Play	Indirect			Renata Fast
2023-11-08	Women - United States	Women - Canada	1	19:52	5	5	0	0	Women - Canada	Renata Fast	Play	Direct			Emma Maltais
2023-11-08	Women - United States	Women - Canada	1	19:50	5	5	0	0	Women - Canada	Emma Maltais	Incomplete Play	Direct			Marie-Philip Poulin

Figure 5: Overall Network for all Sequences Ending in a Goal or Shot on Net

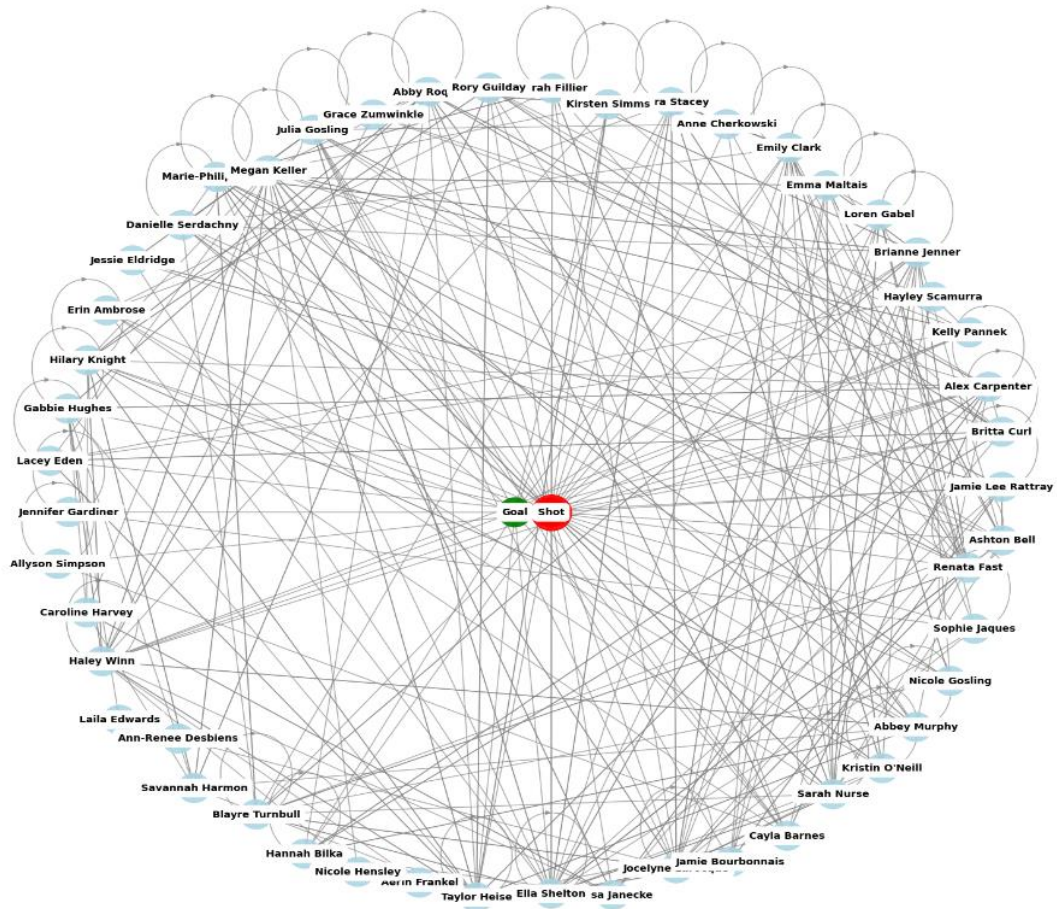


Figure 6: Overall Passing Network Model Output: Alex Carpenter

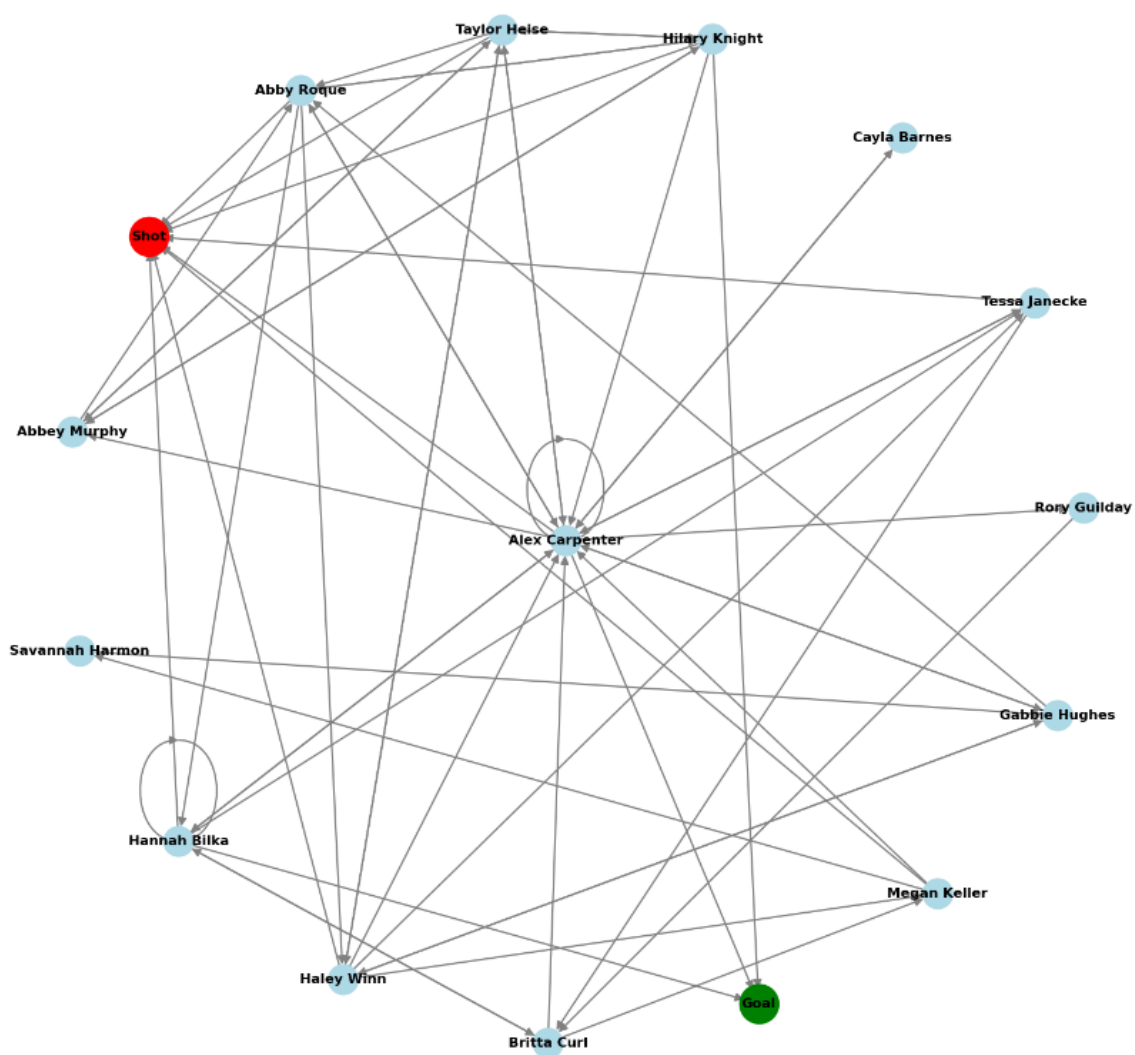


Figure 7: Overall Passing Network Model Output: Taylor Heise

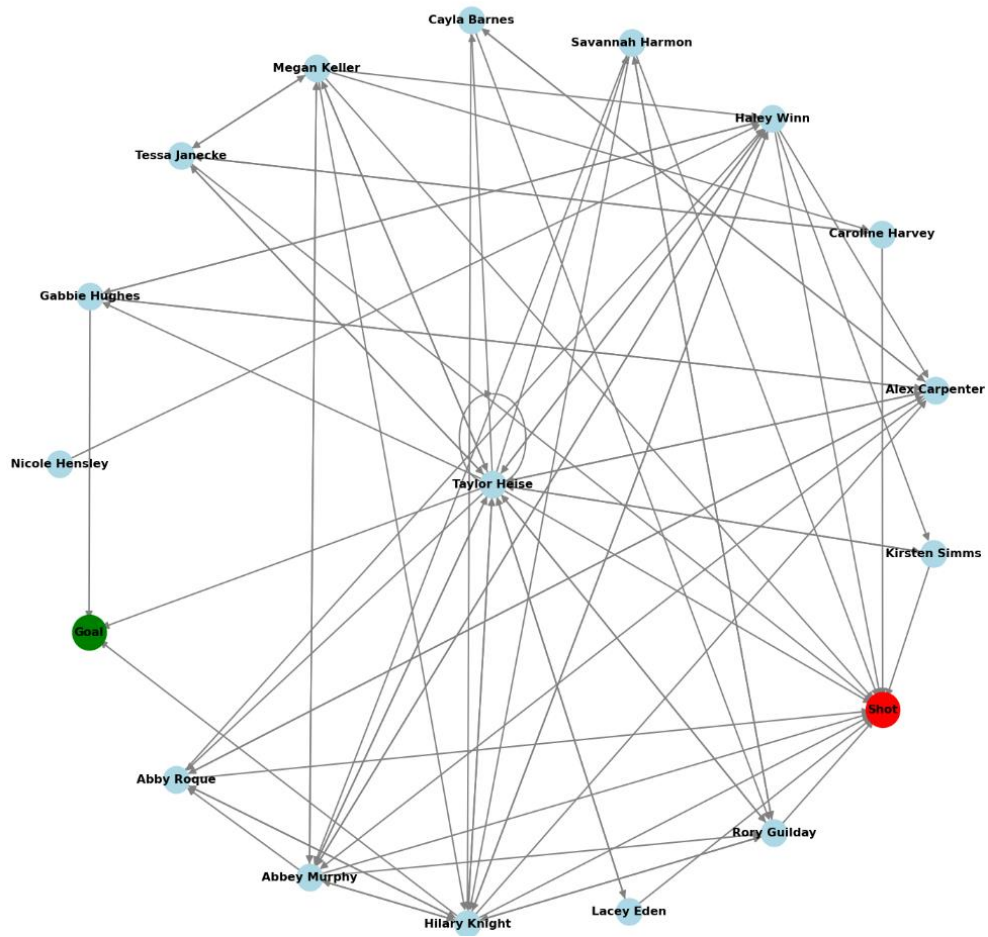


Figure 8: Player Passing Sequences Output from Algorithm: Marie-Philip Poulin

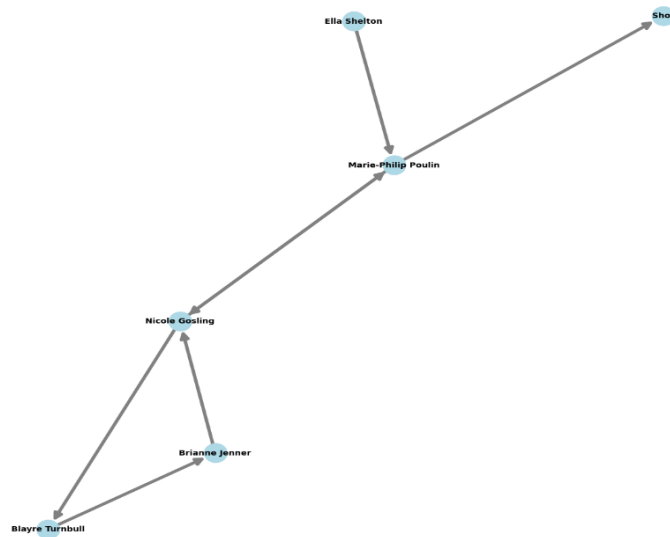


Figure 9: Overall Passing Network Model Output: Savannah Harmon

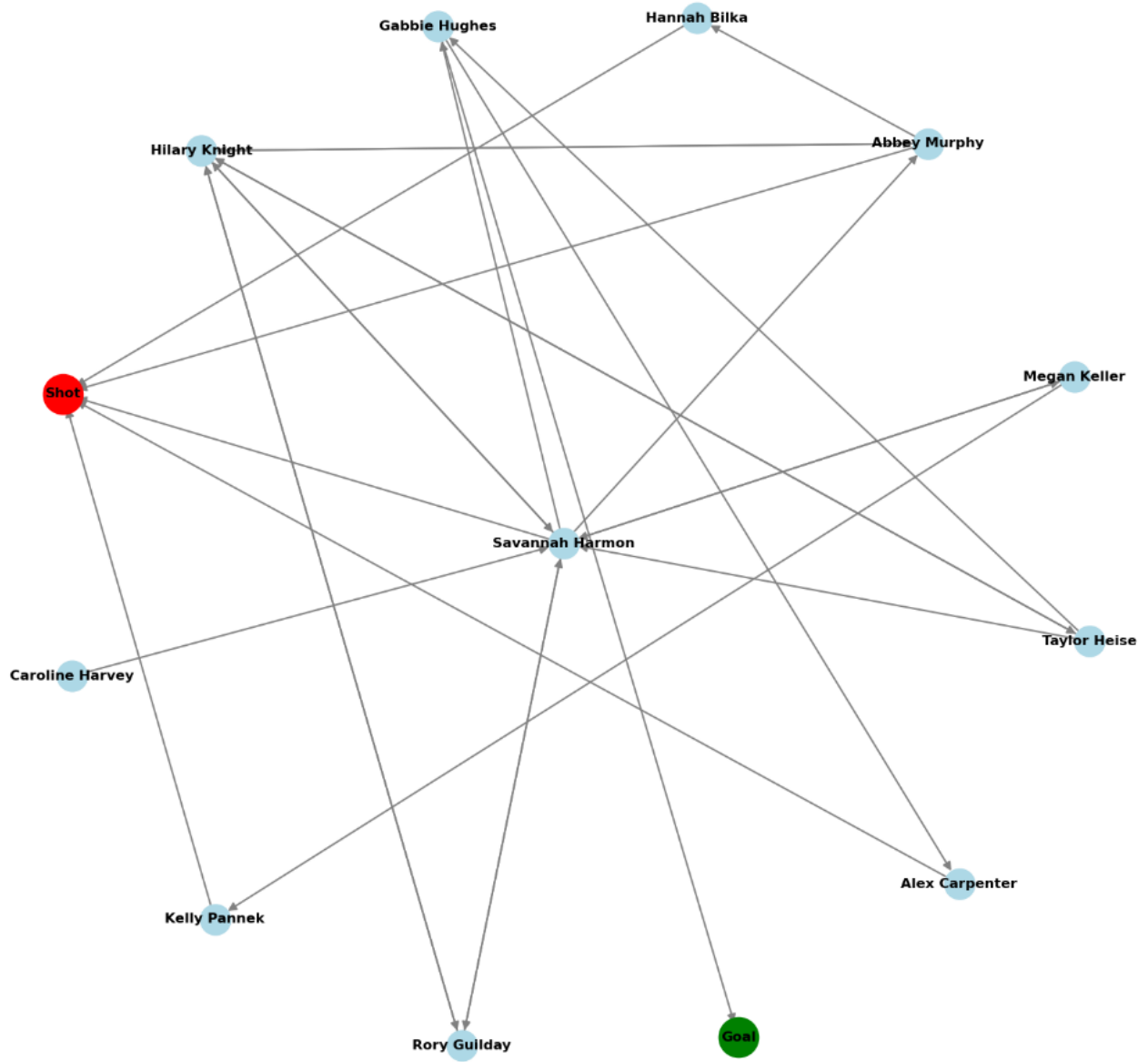


Figure 10: Overall Passing Network Model Output: Renata Fast

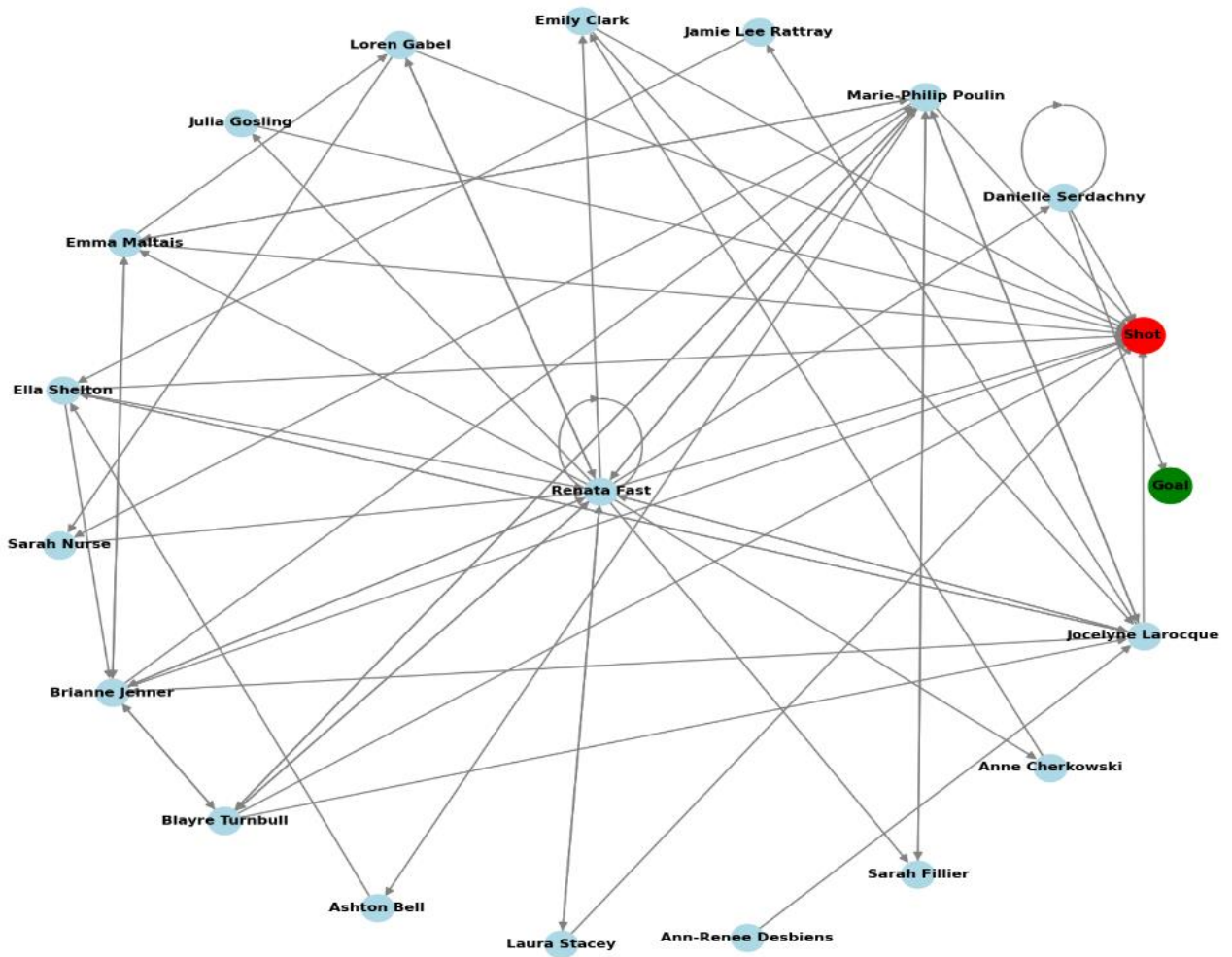


Figure 11: Tableau Dashboard Snippet & Link

